Stochastic Variable Cost and Size Bin Packing with Capacity Loss and Supplier Selection problem

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ISMP 2024

Outline


Problem definition

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- Problem definition
- Two-stage stochastic integer programming model

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- Row aggregation lower bound

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- What's next?

Problem definition I

Supplier selection

- The shipping business has moved from long-term contracts to a more flexible environment based on short-term contracts and "on-spot" transactions
- This shift is rooted into the events that have characterized recent years (pandemic, war, etc.) and impacted logistic systems and supply chains
- The market of shipping has been shaken and the volatility of transportation rates has become normal condition
  - This asks for better decision making but leads also to better profit opportunities
- Shippers have now to frequently select their carrier suppliers and diversify this selection to cover the risk of undelivered capacity in reason of many possible events (mechanical failures, accidents, delays, etc.)

Problem definition I

Supplier selection

- Carrier companies sell storage and transportation services at
  - **Contract rates**
  - **Spot rates**
- Contract rates are applied for medium term purchase agreements for securing future use of capacity and spot rates for just-in-time operations
- **Spot rates** are typically much higher that **contract rates**

Problem definition I

Supplier selection

- We consider the problem of a **shipper** seeking a better profit on operations
- The shipper wants to identify the set of carrier companies to contract with for the next period of operations
- The business setting of the next period is uncertain, the shipper does not known perfectly
  - the amount of orders will be received
  - the actual capacity the carrier companies will actually deliver if contracted
  - the spot rates applied

Problem definition I

Supplier selection

- The shipper looks for the best trade-off of capacity bought at contract rates and spot rates
  - we assume also that the shipper faces a cost for capacity loss in the 2nd-stage and that the concerned suppliers pay a compensation to the shipper
- The shipper wants to compose a portfolio of carrier suppliers that minimizes the expected cost of operations
- The shipper reduces the risk of uncertain factors diversifying the supplier selection

Two-stage stochastic integer programming model I

1st-stage and 2nd-stage decisions

- The shipper's supplier selection problem is two-stage by definition
- The shipper
  - selects the suppliers and buy some capacity at contract rates in the **1st stage** and
  - buy at spot rates the capacity required to ship all orders in the **2nd stage**, when complete information is available

Two-stage stochastic integer programming model I

Sets and notation

- We model the supplier selection problem as an extension of the two-stage modeling of the Stochastic Variable Cost and Size Bin Packing Problem (SVCSBPP) of [3]
- In [3], the focus is on the uncertain 2nd-stage capacity of the bins selected in the 1st-stage from a single supplier (uncertainty is due to possible mechanical failures, delays of delivery, etc.)
- A previous work, i.e., [2], considered the problem of packing an uncertain set of items into a set of bins select in the 1st-stage to resort to the purchase of additional bins in the 2nd-stage
- Neglecting uncertainty, tactical planning problems of transportation are effectively modeled and solved as VCSBPP, e.g., last-mile delivery [4]
- The same problem setting of [2] is considered in [1] form the perspective of e-commerce retailers (such as Amazon or Alibaba) and forecasting machine learning techniques are used to match real-time computational requirements
- Progressive-hedging appears as the best algorithm for two-stage SVCSBPPs, e.g., [1, 2, 3]

Two-stage stochastic integer programming model I

Sets and notation

- Let's see the two-stage stochastic problem formulation

- Given

$\mathcal{M}$: the set of available suppliers

T and $\mathcal{T}$: the bin types available respectively at the 1st-stage and 2nd-stage

$\mathcal{J}_m^t$: the set of bins of type t that the supplier m can provide at the 1st-stage

θ_m: the reliability of supplier m (i.e., the percentage actually delivered on average in the past)

R and ν: the average minimum reliability the shipper targets for selected suppliers and the minimum percentage of capacity to buy at 1st-stage from each selected supplier respectively

Ω: the set of 2nd-stage scenarios

$\mathcal{K}(\omega)$: the set of 2nd-stage bins available under scenario $\omega \in \Omega$

$I_s(\omega)$ and D: the set of order of size d under scenario $\omega \in \Omega$ and the set of possible order sizes respectively

Two-stage stochastic integer programming model I

1st-stage variables

- Given also

  f_m and f_m^t: respectively the cost for contracting supplier m and the contract rate for bins of type t from supplier m

  c_m^t and r_m^t: respectively the cost the shipper face and the compensation the supplier pays for lost capacity (we assume $c_m^t \geq r_m^t$)

  V_m^t, V^τ and $\mathcal{V}_j(\omega)$: respectively the volumes of 1st-stage and 2nd-stage bins according to type t or τ and of the 1st-stage bin j under scenario $\omega \in \Omega$

- 1st-stage variables

  y_m: a binary variable that takes value 1 if supplier $m \in \mathcal{M}$ is selected

  y_{jm}^t: a binary variable that takes value 1 if bin j of type $t \in T$ is bought from supplier $m \in \mathcal{M}$

- and 2nd-stage variables

  x_{dj}: an integer variable counting the number of orders of size d to pack into bin j

  $z_k^\tau(\omega)$: a binary variable that takes value 1 if 2nd-stage bin k of type $\tau \in \mathcal{T}$ is used

Two-stage stochastic integer programming model I

1st-stage problem

We have

$$\min_{y} \sum_{m \in \mathcal{M}} f_m y_m + \sum_{t \in T} \sum_{m \in \mathcal{M}} \sum_{j \in \mathcal{J}_m^t} f_m^t y_{j,m}^t + \mathbb{E}_\xi \left[Q\left(y, \xi(\omega)\right) \right] \tag{1}$$

subject to

$$\sum_{m \in \mathcal{M}} y_m \geq N, \tag{2}$$

$$\sum_{t \in T} \sum_{j \in \mathcal{J}_m^t} V_m^t \; y_{j,m}^t \geq \nu \sum_{t \in T} \sum_{m \in \mathcal{M}} \sum_{j \in \mathcal{J}_m^t} V_m^t \; y_{j,m}^t, \qquad \forall m \in \mathcal{M} \tag{3}$$

$$\sum_{m \in \mathcal{M}} \theta_m \; y_m \geq R \sum_{m \in \mathcal{M}} y_m, \tag{4}$$

$$\sum_{t \in T} \sum_{j \in \mathcal{J}_m^t} y_{j,m}^t \leq y_m, \qquad \forall m \in \mathcal{M}, \tag{5}$$

$$y_{j,m}^t \in \{0,1\}, \qquad \forall t \in T, m \in \mathcal{M}, j \in \mathcal{J}_m^t, \tag{6}$$

$$y_m \in \{0,1\}, \qquad \forall m \in \mathcal{M}, \tag{7}$$

Two-stage stochastic integer programming model I

2nd-stage problem

where

$$Q\left(y, \xi(\omega)\right) = \min_{z(\omega), x(\omega)} \sum_{\tau \in \mathcal{T}} \sum_{k \in \mathcal{K}^\tau(\omega)} g^\tau(\omega) z_k^\tau(\omega) + \\ + \sum_{t \in T} \sum_{m \in \mathcal{M}} \sum_{j \in \mathcal{J}_m^t} (c_m^t - r_m^t)(V^t - \mathcal{V}_j(\omega)) y_{j,m}^t \tag{8}$$

subject to

$$\sum_{j \in \mathcal{J} \cup \mathcal{K}(\omega)} x_{d,j}(\omega) = |I_d(\omega)|, \qquad \forall d \in D, \tag{9}$$

$$\sum_{d \in D} s \cdot x_{d,j}(\omega) \leq \mathcal{V}_j(\omega) y_{j,m}^t, \qquad \forall\, t \in T, m \in \mathcal{M}, j \in \mathcal{J}_m^t, \tag{10}$$

$$\sum_{d \in D} s \cdot x_{d,k}(\omega) \leq V^\tau z_k^\tau(\omega), \qquad \forall\, \tau \in \mathcal{T}, k \in \mathcal{K}^\tau(\omega), \tag{11}$$

$$x_{d,j}(\omega) \in \mathbb{Z}^+, \qquad \forall\, d \in D, j \in \mathcal{J}, \tag{12}$$

$$x_{d,k}(\omega) \in \mathbb{Z}^+, \qquad \forall\, d \in D, k \in \mathcal{K}(\omega), \tag{13}$$

$$z_k^\tau(\omega) \in \{0, 1\}, \qquad \forall\, \tau \in \mathcal{T}, k \in \mathcal{K}^\tau(\omega). \tag{14}$$

Row aggregation of the recourse problem I

- Given the total volume of orders $\Upsilon(\omega) = \sum_{i \in I(\omega)} v_i$ under scenario $\omega \in \Omega$
- We can aggregate constraints (10) and (11) over bins, bin types and suppliers, and consequently discard constraints (9)
- And we obtain recourse problem

$$Q\left(y, \xi(\omega)\right)$$

subject to

$$\Upsilon(\omega) \leq \sum_{\substack{m \in \mathcal{M}, \\ t \in T, \\ j \in \mathcal{J}_m^t}} \mathcal{V}_j(\omega) y_{j,m}^t + \sum_{\substack{\tau \in \mathcal{T}, \\ k \in \mathcal{K}^{\tau,s}}} V^\tau z_k^\tau(\omega), \qquad \forall\, \omega \in \Omega, \tag{15}$$

$$z_k^\tau(\omega) \in \{0,1\}, \qquad \forall\, \tau \in \mathcal{T}, k \in \mathcal{K}^\tau(\omega), \tag{16}$$

Row aggregation of the recourse problem I

- By solving problem

$$\min_{y} \sum_{m \in \mathcal{M}} f_m y_m + \sum_{t \in T} \sum_{m \in \mathcal{M}} \sum_{j \in \mathcal{J}_m^t} f_m^t y_{j,m}^t + \mathbb{E}_{\xi} \left[Q\left(y, \xi(\omega)\right) \right]$$

subject to

$$(2),\ (3),\ (4),\ (5),\ (6),\ \text{and}\ (15)$$

$$\begin{aligned} y_{j,m}^t &\in \{0,1\}, && \forall t \in T, m \in \mathcal{M}, j \in \mathcal{J}_m^t, \\ y_m &\in \{0,1\}, && \forall m \in \mathcal{M}, \\ z_k^\tau(\omega) &\in \{0,1\}, && \forall\, \tau \in \mathcal{T}, k \in \mathcal{K}^\tau(\omega), \end{aligned}$$

we obtain a **valid lower bound** of the problem

- Note that we discarded all scenario-dependent packing variables $x_{dj}(\omega)$

- The obtained solution of 1st-stage variables y_m and y_{jm}^t can then be evaluated in the original problem for a set of scenarios $\tilde{\Omega} \subset \Omega$
  - This step provides an **upper bound** for the **recourse problem** (RP) evaluation

Evaluation deterministic-equivalent ILPs I

- We evaluate the deterministic-equivalent ILPs of the two-stage problem
  - (1)-(14) and
  - that of the row aggregation lower bound

- The deterministic-equivalent ILP of (1)-(14) can be solved only for a limited number of scenarios (5, 10, 20)
  - Let S be the set of generated scenario approximating Ω

- We generate sets of instances of small size 100-200 orders per scenario and 3 suppliers

- To speed up the model resolution we implemented a procedure for the complete enumeration of supplier selections (this is possible due to small number of suppliers of an instance)

Instance generation I

Generated instances reflect the following (relevant) characteristics

- Three suppliers, $|\mathcal{M}| = 3$
- Select at least two suppliers, $N = 2$, and a minimum volume of bin capacity of 20%, $\nu = 0.2$
- With minimum ave. reliability of 80%, $R = 0.8$
- Reliability of each supplier θ_m drawn from $U(0.7, 0.9)$
- Nr. of items per scenario drawn from $U(100, 200)$
- Three 1st-stage bin types with volumes $V_t \in \{\, 50,\ 100,\ 150 \,\}$
- Nr. of 2st-stage bins per bin type $|\mathcal{J}_m^t| = \lceil \frac{\textbf{MTVIS}}{V^t} \rceil$; MTVIS is the maximum total volume of items over the generated scenarios S
- Nr. of 1st-stage bins per type $|\mathcal{K}_m^t|$ drawn from $U_{\textbf{int}}(0.1|\mathcal{J}_m^t|, 0.5|\mathcal{J}_m^t|)$

Instance generation I

- Reliability of supplier m under scenario $s \in S$ is computed as $\tilde{\theta}^s_m = \theta_m + \varepsilon$ with ε drawn from
  - $U(-0.1, 0.1)$, **low variance** (LV), or
  - $U(-0.3, 0.3)$, **high variance** (HV)

- Aggregated capacity loss of supplier m under scenario $s \in S$ is computed as $\mathcal{L}^s_m = (1 - \tilde{\theta}^s_m) \sum_{i \in I_s} \nu^s_i$

- 2nd-stage loss $\mathcal{V}^s_j$ of 1st-stage bin j under scenario $s \in S$, is computed according to two cases

  Uniform: $\mathcal{V}^s_j = V^t_m - l^s_m$ with $l^s_m = \frac{\mathcal{L}^s_m}{|\mathcal{J}^t_m|}$

  Localized: $\mathcal{V}^s_j$ is 0 or V^t_m such that $\sum_{j \in \mathcal{J}^t} \mathcal{V}^s_j + \mathcal{L}^s_m = V^t_m |\mathcal{J}^t|$

Instance generation I

- 1st-stage costs
  - Contract cost $f_m = \zeta_m \cdot \nu \cdot \max_{s \in S} \sum_{i \in I_s} \nu_i^s$ with $\zeta_m \sim U(0.1, 0.5)$
  - Contract rates $f_m^t = V^t(1 + \gamma_m^t)$ with $\gamma_m^t \sim U(-0.25, 0.25)$

- 2nd-stage costs
  - Capacity loss cost $c_m^t = 1.25 \frac{f_m^t}{V^t}$ and compensation $r_m^t = \frac{f_m^t}{V^t}$
  - Spot rates $g_m^t = f_m^t(1 + \beta_m^t)$ with
    - $\beta_m^t \sim U(0.25, 0.5)$, **small spread** (SS), or
    - $\beta_m^t \sim U(0.5, 0.75)$, **large spread** (LS)

Numerical experiments I

Expected Value of Perfect Information, EVPI, and Value of Stochastic Solution, VSS

- 20 instances for each set, 20 or 120 scenarios for each instance
- All four combination of low/high variance of supplier reliability (LV/HV) and small/large spread of spot rates (SS/LS) w.r.t. contract rates
- For the WS and EEV computation, the model w/ packing vars does not always solve to optimality the single-scenario problem, but the gap is limited ($< 1\%$)
  - if not optimal, the best bound is taken
  - the same for the RP objective function evaluation
- All the experiments are run as single thread processes on a PC equipped with a 12th Gen Intel Core i9-12900 3.8GHz, with 32GB of RAM and running Ubuntu 22.04 LTS
- Time-limits are
  - 600 sec. for each supplier selection of the complete enumeration procedure, when there is no scenario separation (i.e., EV and RP computation)
  - 30 sec. for each supplier selection in the case of scenario separation (i.e., WS and EEV computation)

Numerical experiments I

Expected Value of Perfect Information, EVPI, and Value of Stochastic Solution, VSS

Model with packing variables (PV) and model row aggregation (RA)

Model	Loss type	Instance set	EVPI	VSS	RP gap
PV	U	LV-SS	2.14%	**8.13%**	1.75%
PV	U	LV-LS	5.18%	3.58%	3.98%
PV	U	HV-SS	4.24%	**8.58%**	1.43%
PV	U	HV-LS	**7.21%**	5.36%	4.87%
PV	L	LR-SS	4.98%	3.58%	4.25%
PV	L	LR-LS	**7.64%**	1.98%	3.81%
PV	L	HR-SS	4.81%	3.83%	3.14%
PV	L	HR-LS	**8.43%**	2.22%	3.08%
RA	U	LV-SS	2.35%	**7.84%**	0.18%
RA	U	LV-LS	5.60%	3.15%	0.24%
RA	U	HV-SS	4.42%	**8.31%**	0.11%
RA	U	HV-LS	**7.68%**	4.79%	0.33%
RA	L	LR-SS	5.60%	2.70%	0.00%
RA	L	LR-LS	**8.33%**	1.21%	0.00%
RA	L	HR-SS	5.52%	2.97%	0.00%
RA	L	HR-LS	**9.09%**	1.49%	0.00%

Table: EVPI and VSS evaluation, 20 instances for each set, 20 scenarios; uniform loss (U) or localized loss (L)

Numerical experiments I

Expected Value of Perfect Information, EVPI, and Value of Stochastic Solution, VSS

- We evaluate the quality of the RP solution we obtain if we inject the RA model solution of 1st-stage variable into model PV
  - Note that by doing so we have separation of scenarios and a much faster evaluation

- The evaluation is done by comparing this RP solution and that of model PV as the gap w.r.t. the best RP bound (PV or RA)

Loss	Set	Best RP bound	RP of PV	Gap	RP injected sol.	Gap w/ best
U	LV-SS	3094.7	3146.6	**1.68%**	3157.2	2.02%
U	LV-LS	3370.2	3503.9	3.97%	3402.5	0.96%
U	HV-SS	3165.2	3210.4	**1.43%**	3232.5	2.13%
U	HV-LS	3225.9	3384.7	4.92%	3260.7	**1.08%**
L	LV-SS	2782.9	2899.7	4.20%	2810.2	**0.98%**
L	LV-LS	2948.1	3060.8	3.82%	2976.9	**0.98%**
L	HV-SS	2815.0	2903.4	3.14%	2841.7	**0.95%**
L	HV-LS	2827.4	2915.1	3.10%	2851.9	**0.87%**

Table: RP evaluation, model with packing variables (PV) *vs.* injecting row aggregation (RA) 1-stage solution

Numerical experiments I

Expected Value of Perfect Information, EVPI, and Value of Stochastic Solution, VSS

- We evaluate also the row aggregation model for a larger set of scenarios; 120 scenarios

Loss type	Instance set	EVPI	VSS	RP gap
U	LVRLSR	2.17%	**9.47%**	0.15%
U	LVRHSR	5.73%	5.52%	0.40%
U	HVRLSR	4.74%	**8.92%**	0.18%
U	HVRHSR	**7.57%**	6.45%	0.47%
L	HVRHSR	**8.84%**	1.80%	0.00%
L	HVRLSR	**5.87%**	2.52%	0.00%
L	LVRHSR	**8.59%**	1.76%	0.00%
L	LVRLSR	**5.80%**	2.98%	0.00%

Table: EVPI and VSS evaluation, 120 scenarios; uniform loss (U) localized loss (L)

Numerical experiments I

Some considerations

- Instances with uniform loss of bin capacity seems to be slightly harder to solve
- For tested instances, characterized by a small amount of items, the EVPI and VSS indicators suggest a relevant impact of stochastic optimization for the problem
- The large number of ILP variables due to the several suppliers to select makes the problem hard to solve, meta-heuristic approaches reducing the solution space can be effective
- The tested procedure of 1st-stage variables solution injection suggests as promising stochastic solution algorithms allowing separation of scenarios (e.g., progressive-hedging)

Looking the way forward I

- Devise a meta-heuristic solution approach for the selection of suppliers
- Devise a pseudo-exact progressive-hedging algorithm for solving the recourse problem

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